CPSC 57700 Machine Learning Operations

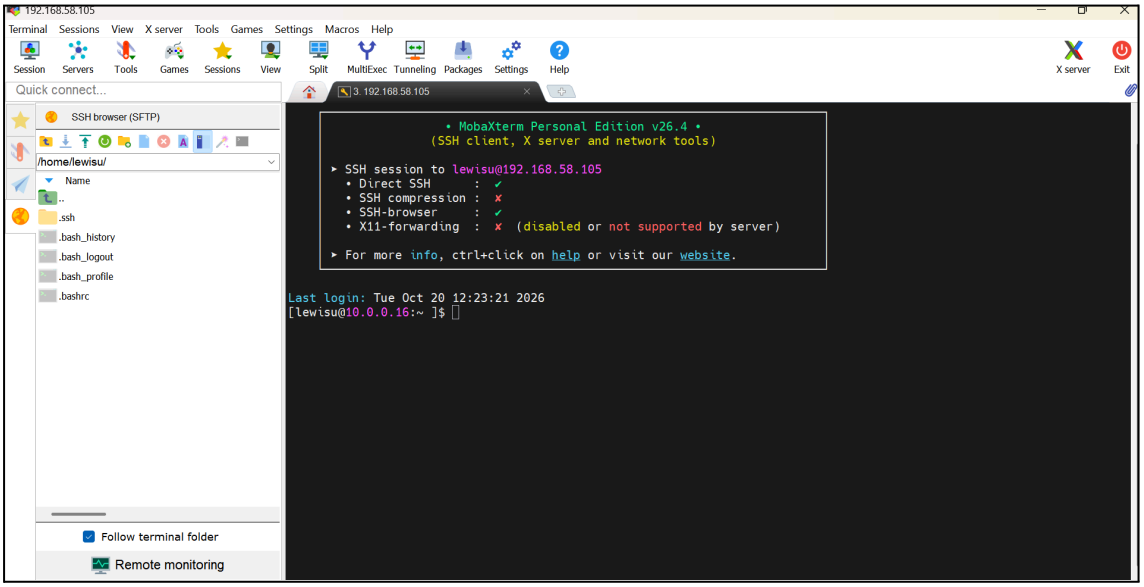
Fall 2026 -Week 1 Assignment

INFO

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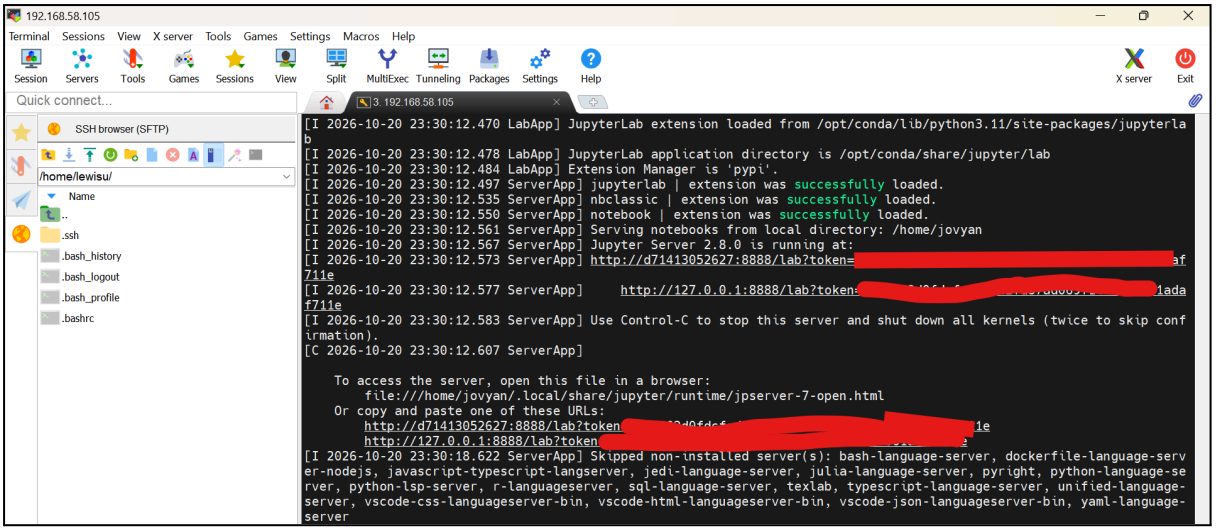
Part 1: Install and configure Linux VM

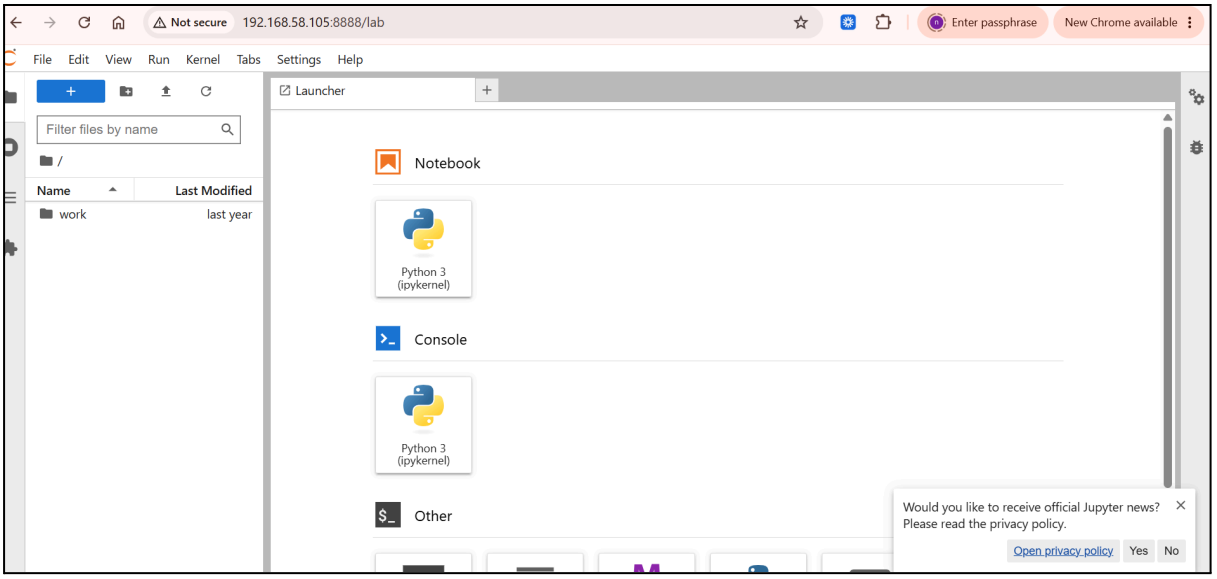
9. Start the VM and ssh to it using the ssh client



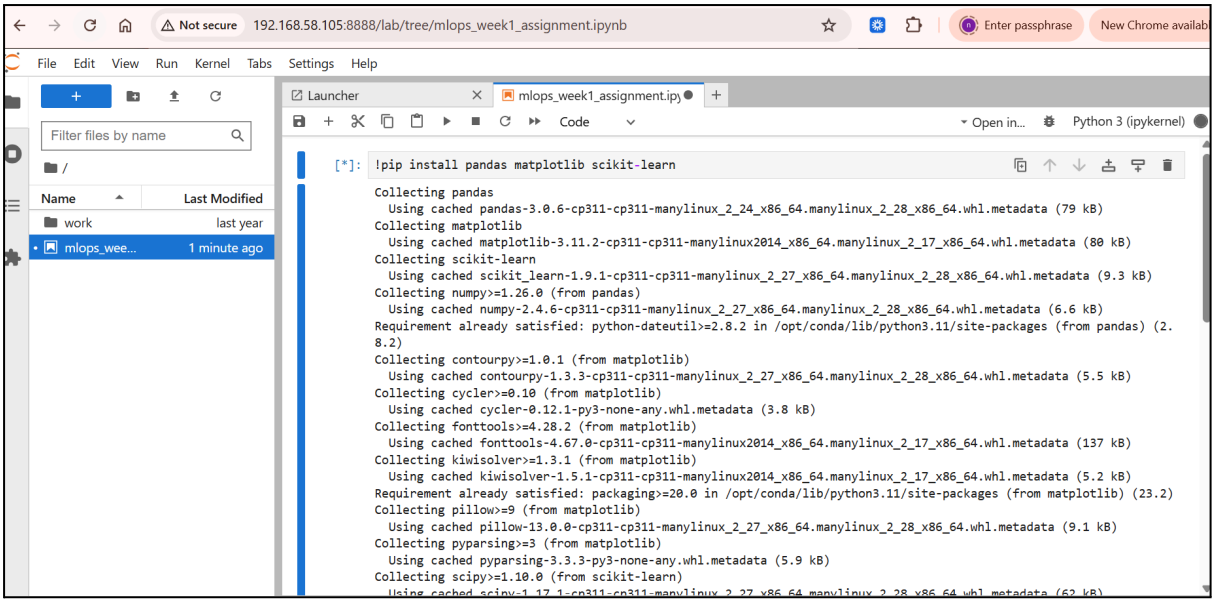
Part 2: Create an ML workbench

5. The messages displayed towards the end will contain a URL with a long token. You have to use this URL but replace the hostname with the IP address of the VM. For example, this is what you should use. Replace the highlighted part with values from your setup.



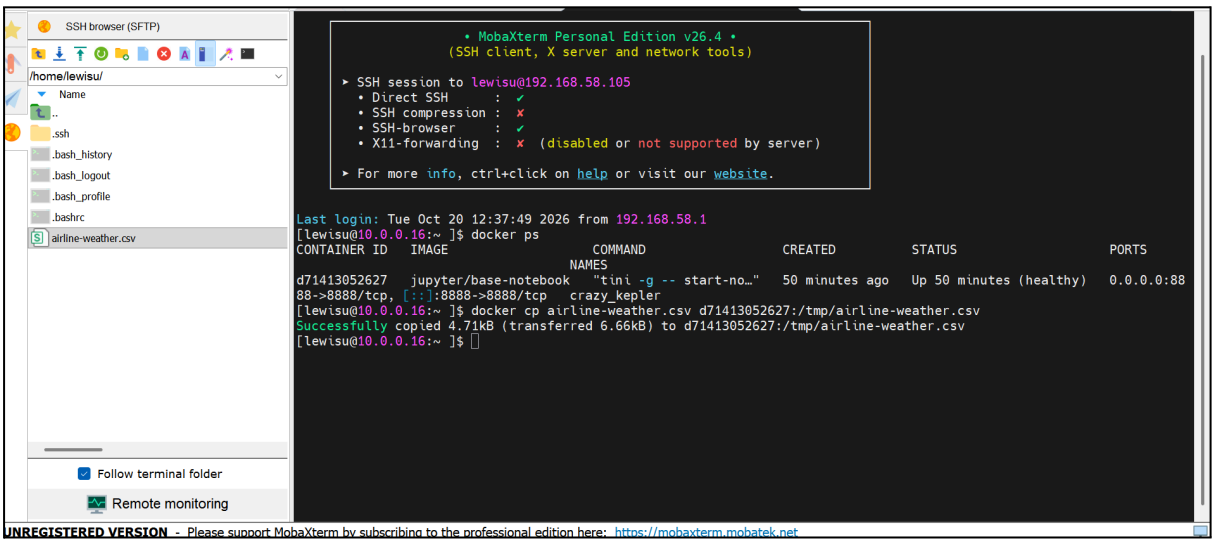


6. In an empty cell, use the pip command to install the following python packages a. Pandas b. matplotlib c. scikit-learn



Part 3: Use sample data and build an ML model

2. Copy the csv file into the Jupyter container's /tmp folder by using the "docker cp" command



3. Write python code inside the Jupyter notebook to

a. Load the data from the CSV file
b. Use Pandas, sklearn, matplotlib
c. Convert text entries into numeric entries in your python code
d. Algorithm to use: RandomForestClassifier
e. Use the following features as inputs: i. Weather, CrewAvailable, BusyRunways
f. Use the following output variable i. DelayedYN

g. Use the scikit-learn train-test split
h. Train the ML model
i. Predict the values using the test data
j. Compute the metrics such as Accuracy, F1 etc.. to gauge the performance of the model
k. Plot the important features
l. Display metrics (accuracy, feature importance etc..)

```
Accuracy: 0.7777777777777778
F1 Score: 0.8461538461538461

 Classification Report:
              precision    recall  f1-score   support

           0       0.60      0.60      0.60         5
           1       0.85      0.85      0.85        13

    accuracy                           0.78        18
   macro avg       0.72      0.72      0.72        18
weighted avg       0.78      0.78      0.78        18


 Features Importance:
Weather          0.728665
BusyRunways      0.212617
CrewAvailable    0.058718
dtype: float64
```

